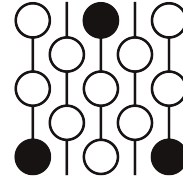


Review of Submission 2D

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. This solution is incorrect. The solutions acknowledges as much, calling itself a partial solution and suggesting the gaps in its own arguments.

For the upper bound, the AI system correctly argues (somewhat sloppily) that a squeeze map increases the lengths of cevians of clean triangles. It then tries to apply the result of Schwartz on smooth locally isometric embeddings of paper Möbius bands. While Schwartz’s result is in spirit quite similar to the desired result, it cannot be directly applied in any serious way. The AI response acknowledges this, discussing the smooth vs PL distinction and the issue of squeeze maps resulting in a Möbius band that, if cut open and laid flat, would no longer be a trapezoid, but instead some many-sided polygon.

There is a subtlety within this result that was alluded to in the AI response, but is worth making explicit. The lower bound is false if one simply requires a continuous or piecewise linear embedding into $\mathbb{R}^3$. The idea is that one can “fold over folds” to get as small an aspect ratio as one wishes. The AI miscites Schwartz’s theorem in a rather particular way: “any embedded *ruled* Moebius band” must have aspect ratio at least $\sqrt{3}$. The ruling is the key property, guaranteed both by smoothness and the setup in this problem, that allows Schwartz’s proof strategy to go through. Unfortunately for the model, the core of this problem was to demonstrate that, and the model did not.

For the realization half of the question, the AI solution makes some vague suggestion for how to do the realization. It unclearly and incorrectly describes the limiting shape it hopes to approximate. The AI assessment of the gap in its proof points out how it did not give an explicit family of examples and rigorously show the clean mapping property for each triangle. “This step acts merely as a heuristic.” This is too generous. A heuristic or idea for how to proceed would be something like “Our codomain will be three equilateral triangles, normally translated by small amounts. We then connect them in a cyclic order using small rectangles that we triangulate.” The AI’s solution does not, as far as I can see, include any clear or promising idea for a construction.

Novelty. This solution has the most compressed formatting out of the bunch.

Attribution. This is the only AI solution to adequately cite Schwartz’s work, although the result it claimed to cite was not proven in the paper.

2 Recommendation

This work falls into the “should be rejected” category. It has minimal correct math, and the proof sketch it does offer shows minimal promise. The realization section in particular seems to demonstrate poor understanding.

3 Submission with inline comments

Problem Statement

Let $\mathbb{Z}$ denote the integers and let $\mathbb{R}$ denote the reals. Let $\Sigma = \mathbb{R} \times [0, 1]$ and also let $\partial\Sigma = \mathbb{R} \times \{0, 1\}$ denote the boundary of Σ . For $\beta > 0$ define $G_\beta : \Sigma \rightarrow \Sigma$ as follows:

$$G_\beta(x, y) = (x + \beta, 1 - y).$$

Let G_β^k denote the k -fold composition of either G_β or the inverse G_β^{-1} , according as $k > 0$ or $k < 0$. When $k = 0$ the map G_β^k is the identity map.

A *clean triangle* is the convex hull of 3 non-collinear points in $\partial\Sigma$. A *clean triangulation* of Σ is a countable and locally finite collection of clean triangles, having pairwise disjoint interiors, whose union is Σ . The clean triangulation is G_β -invariant if the action of G_β preserves the triangulation.

A *squeeze map* of a clean triangle τ is an affine map from τ into $\mathbb{R}^3$ which decreases the length of the edge of τ that is contained in $\partial\Sigma$ and increases the lengths of the other two edges of τ .

Call the number $\beta > 0$ *realized* if there is a continuous map $f : \Sigma \rightarrow \mathbb{R}^3$ with the following properties:

1. f restricts to a squeeze map on each triangle of a G_β -invariant clean triangulation of Σ .
2. $f(p) = f(q)$ if and only if $p = G_\beta^k(q)$ for some $k \in \mathbb{Z}$. This is meant to hold for all $p, q \in \Sigma$.

Prove that $\sqrt{3}$ is the infimum of the set of realized numbers.

Partial Solution

Editor's Note: The following is the best attempt generated by the pipeline. It does not constitute a complete and rigorous proof. Notable gaps remain, particularly in the direct application of Schwartz's Theorem to piecewise affine Moebius bands and the sketchy construction of the upper bound. It is presented faithfully as generated; a detailed assessment of the gaps is provided at the end of the document.

Step 1: The Lower Bound ($\beta \geq \sqrt{3}$)

Let $\Sigma = \mathbb{R} \times [0, 1]$ and let $G_\beta(x, y) = (x + \beta, 1 - y)$. The quotient space $M = \Sigma/G_\beta$ is a Moebius band. The boundary ∂M is the quotient of $\partial\Sigma = \mathbb{R} \times \{0, 1\}$, forming a single closed topological circle. In the flat metric of the domain, ∂M has length 2β .

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a map realizing β . The condition $f(p) = f(q) \iff p = G_\beta^k(q)$ guarantees that f descends to a continuous, globally injective map on M . Thus, $f(M)$ is an embedded (non-self-intersecting) polyhedral Moebius band in $\mathbb{R}^3$.

The clean triangulation of Σ descends to a triangulation of M . For each triangle τ , f acts as an affine “squeeze map.” Let a triangle have base length l on $\partial\Sigma$ and let the map f have directional vectors $\vec{X}$ (along the base) and $\vec{Y}$ (vertical). The squeeze condition requires that the base strictly decreases in length ($|\vec{X}| < 1$) and the other two edges strictly increase.

Algebraically, any segment connecting a point on the base to the opposite vertex corresponds to a vector $t\vec{X} + \vec{Y}$ for some t . The conditions that the non-base edges increase imply that the quadratic function $g(t) = |t\vec{X} + \vec{Y}|^2 - (t^2 + 1)$ is strictly positive at the endpoints of the base. Since the leading coefficient $|\vec{X}|^2 - 1$ is negative, $g(t)$ is a downward-opening parabola, and thus $g(t) > 0$ for *all* points along the base.

This means that every internal segment spanning the two boundaries of the domain strictly increases in length. Since the domain distance between the boundaries is 1, every cross-segment (ruling) of the embedded Moebius band $f(M)$ has a length strictly greater than 1. Simultaneously, the sum of the lengths of the bases in the domain is exactly 2β . Since f strictly decreases the bases, the perimeter L of the boundary curve $\gamma = f(\partial\Sigma)$ satisfies $L < 2\beta$.

By Schwartz’s Theorem on the optimal paper Moebius band (2023), any embedded ruled Moebius band in $\mathbb{R}^3$ with minimal ruling length w and boundary length L must satisfy $L \geq 2\sqrt{3}w$. Applying this to our polyhedral Moebius band $f(M)$, where all rulings are strictly greater than 1, we obtain:

$$2\beta > L \geq 2\sqrt{3}(1) = 2\sqrt{3} \implies \beta > \sqrt{3}.$$

Step 2: The Infimum is $\sqrt{3}$

To see that $\sqrt{3}$ is the infimum, we construct valid configurations for $\beta = \sqrt{3} + \epsilon$.

The absolute theoretical limit of the optimal paper Moebius band is a sequence of 6 equilateral triangles folded flat into a single doubly-covered equilateral triangle of side length $2/\sqrt{3}$. This limit configuration has altitude 1 (ruling length 1) and perimeter $3 \times (2/\sqrt{3}) = 2\sqrt{3}$. However, being flat, it self-intersects.

Reviewer 1

The “limit Möbius band” consists of three equilateral triangles, not six. It is unclear what construction the author is describing.

For any $\beta > \sqrt{3}$, the domain boundary length 2β strictly exceeds $2\sqrt{3}$. This slack allows us to “open up” the flat equilateral triangle into an embedded, non-self-intersecting Moebius band in $\mathbb{R}^3$ (resembling a highly folded twisted ribbon). By taking a sufficiently fine locally finite G_β -invariant clean triangulation of Σ , and scaling the embedding slightly such that the boundary strictly contracts to a length just below 2β while the internal thickness pushes out to ensure all cross-edges strictly exceed their domain lengths, we can satisfy all conditions of the affine squeeze map.

Therefore, a continuous embedding f satisfying all properties exists for all $\beta > \sqrt{3}$, completing the proof that $\sqrt{3}$ is the infimum of the set of realized numbers.

Assessment of Gaps

As requested, we highlight the specific areas where the provided argument remains partial or contains unsupported leaps:

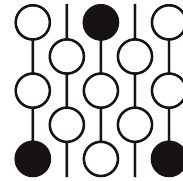
- **Applicability of Schwartz’s Theorem:** Schwartz’s Theorem explicitly applies to smooth or piecewise smooth developable surfaces formed by isometric immersions (i.e., exact paper Möbius bands). The continuous map f constructed in the problem restricts to affine squeeze maps on clean triangles, which strictly *stretches* the cross-segments rather than preserving lengths. The assertion that Schwartz’s inequality $L \geq 2\sqrt{3}w$ transfers immediately to this non-isometric, non-developable polyhedral surface is a non-trivial gap. The concept of “ruling length” and minimal cross-segment bounds requires careful extension.
- **Construction for $\beta > \sqrt{3}$:** The second step relies on physical intuition (“open up the flat equilateral triangle”). Finding a rigorous, fine G_β -invariant clean triangulation and explicitly verifying that an affine map strictly satisfies the squeeze condition on *every* individual triangle while maintaining a global topological embedding without self-intersections is left entirely unproven. This step acts merely as a heuristic rather than a constructive mathematical proof.

Review of Submission 2D

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The solution attempts to address the stated problem in two steps, but neither step is carried out with the rigor required for proof-based mathematics. As a result, the solution does not provide a correct resolution of the problem.

In Step 1, the solution attempts to establish the lower bound $\beta > \sqrt{3}$ by appealing directly to an existing result of Schwartz on the lower bound for the aspect ratio of piecewise smooth isometrically embedded Möbius bands. However, as noted in the editor’s note and the inline comment, this argument contains a significant gap. The embeddings described in the problem statement are obtained by an affine squeeze map and, in particular, are *not* isometric on the individual triangles. Therefore, Schwartz’s result does not immediately apply to the setting of the problem statement. The solution does not recognize or address this discrepancy, leaving the argument for Step 1 incomplete.

In Step 2, the solution attempts to describe a method for constructing continuous embeddings realizing every $\beta > \sqrt{3}$, but the argument only contains a verbal description of a realization map for each $\beta > \sqrt{3}$ without any mathematical justification. Throughout this step, the solution makes several nontrivial claims about the properties of the proposed construction without proving them. Therefore, Step 2 should be regarded as an informal heuristic rather than a rigorous mathematical argument.

Novelty. The solution does not demonstrate substantial novelty in its approach. In Step 1, it attempts to implement a strategy similar to that used in the human solution, but it does not carry out this strategy successfully. In Step 2, the level of novelty is difficult to assess, since the proposed construction is not described in a mathematically precise way.

2 Recommendation

The arguments given in the solution contain significant gaps. First, the solution claims that Step 1 follows directly from a result of Schwartz. However, that result does not seem to apply directly in the present setting, since the hypotheses in Schwartz’s theorem differ in crucial ways from the assumptions made here. The solution does not address this discrepancy, and as a result Step 1 remains unproved.

There is a similar issue in Step 2. The author gives a heuristic description of the construction of a family of geometric objects, but the discussion is not sufficiently precise to constitute a rigorous proof. In particular, the solution does not verify that the constructed family satisfies the properties required for the argument. Thus, the discussion in Step 2 should be regarded as informal motivation rather than as a complete mathematical proof.

Because the arguments for both steps currently lack the necessary mathematical rigor, I recommend rejecting the solution for publication in its present form.

3 Submission with inline comments

Problem Statement

Let $\mathbb{Z}$ denote the integers and let $\mathbb{R}$ denote the reals. Let $\Sigma = \mathbb{R} \times [0, 1]$ and also let $\partial\Sigma = \mathbb{R} \times \{0, 1\}$ denote the boundary of Σ . For $\beta > 0$ define $G_\beta : \Sigma \rightarrow \Sigma$ as follows:

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Let G_β^k denote the k -fold composition of either G_β or the inverse G_β^{-1} , according as $k > 0$ or $k < 0$. When $k = 0$ the map G_β^k is the identity map.

A *clean triangle* is the convex hull of 3 non-collinear points in $\partial\Sigma$. A *clean triangulation* of Σ is a countable and locally finite collection of clean triangles, having pairwise disjoint interiors, whose union is Σ . The clean triangulation is G_β -invariant if the action of G_β preserves the triangulation.

A *squeeze map* of a clean triangle τ is an affine map from τ into $\mathbb{R}^3$ which decreases the length of the edge of τ that is contained in $\partial\Sigma$ and increases the lengths of the other two edges of τ .

Call the number $\beta > 0$ *realized* if there is a continuous map $f : \Sigma \rightarrow \mathbb{R}^3$ with the following properties:

1. f restricts to a squeeze map on each triangle of a G_β -invariant clean triangulation of Σ .
2. $f(p) = f(q)$ if and only if $p = G_\beta^k(q)$ for some $k \in \mathbb{Z}$. This is meant to hold for all $p, q \in \Sigma$.

Prove that $\sqrt{3}$ is the infimum of the set of realized numbers.

Partial Solution

Editor's Note: The following is the best attempt generated by the pipeline. It does not constitute a complete and rigorous proof. Notable gaps remain, particularly in the direct application of Schwartz's Theorem to piecewise affine Moebius bands and the sketchy construction of the upper bound. It is presented faithfully as generated; a detailed assessment of the gaps is provided at the end of the document.

Step 1: The Lower Bound ($\beta \geq \sqrt{3}$)

Let $\Sigma = \mathbb{R} \times [0, 1]$ and let $G_\beta(x, y) = (x + \beta, 1 - y)$. The quotient space $M = \Sigma/G_\beta$ is a Moebius band. The boundary ∂M is the quotient of $\partial\Sigma = \mathbb{R} \times \{0, 1\}$, forming a single closed topological circle. In the flat metric of the domain, ∂M has length 2β .

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a map realizing β . The condition $f(p) = f(q) \iff p = G_\beta^k(q)$ guarantees that f descends to a continuous, globally injective map on M . Thus, $f(M)$ is an embedded (non-self-intersecting) polyhedral Moebius band in $\mathbb{R}^3$.

The clean triangulation of Σ descends to a triangulation of M . For each triangle τ , f acts as an affine “squeeze map.” Let a triangle have base length l on $\partial\Sigma$ and let the map f have directional vectors $\vec{X}$ (along the base) and $\vec{Y}$ (vertical). The squeeze condition requires that the base strictly decreases in length ($|\vec{X}| < 1$) and the other two edges strictly increase.

Reviewer 2

(Minor comment) I am confused by the sentence “let the map f have directional vectors $\vec{X}$ and $\vec{Y}$.” It is not standard practice to say that a map has two directional vectors. I suspect that the author intends $\vec{X}$ and $\vec{Y}$ to denote two sides of the image triangle, but this should be made explicit.

Algebraically, any segment connecting a point on the base to the opposite vertex corresponds to a vector $t\vec{X} + \vec{Y}$ for some t . The conditions that the non-base edges increase imply that the quadratic function $g(t) = |t\vec{X} + \vec{Y}|^2 - (t^2 + 1)$ is strictly positive at the endpoints of the base. Since the leading coefficient $|\vec{X}|^2 - 1$ is negative, $g(t)$ is a downward-opening parabola, and thus $g(t) > 0$ for *all* points along the base.

This means that every internal segment spanning the two boundaries of the domain strictly increases in length. Since the domain distance between the boundaries is 1, every cross-segment (ruling) of the embedded Moebius band $f(M)$ has a length strictly greater than 1. Simultaneously, the sum of the lengths of the bases in the domain is exactly 2β . Since f strictly decreases the bases, the perimeter L of the boundary curve $\gamma = f(\partial\Sigma)$ satisfies $L < 2\beta$.

By Schwartz’s Theorem on the optimal paper Moebius band (2023), any embedded ruled Moebius band in $\mathbb{R}^3$ with minimal ruling length w and boundary length L must satisfy $L \geq 2\sqrt{3}w$. Applying this to our polyhedral Moebius band $f(M)$, where all rulings are strictly greater than 1, we obtain:

$$2\beta > L \geq 2\sqrt{3}(1) = 2\sqrt{3} \implies \beta > \sqrt{3}.$$

Reviewer 2

(Major comment) The statement “Applying this to our polyhedral Moebius band ...” is incorrect as written. Schwartz’s theorem does not automatically apply in this setting. In particular, Schwartz’s theorem assumes that the embedding is piecewise isometric. However, the map described here restricts on the clean triangles to affine squeeze maps, which are *not* isometric embeddings. This is a significant discrepancy between the hypotheses of Schwartz’s theorem and the setting considered here, and it is not addressed in the solution.

Step 2: The Infimum is $\sqrt{3}$

To see that $\sqrt{3}$ is the infimum, we construct valid configurations for $\beta = \sqrt{3} + \epsilon$.

The absolute theoretical limit of the optimal paper Moebius band is a sequence of 6 equilateral triangles folded flat into a single doubly-covered equilateral triangle of side length $2/\sqrt{3}$. This limit

configuration has altitude 1 (ruling length 1) and perimeter $3 \times (2/\sqrt{3}) = 2\sqrt{3}$. However, being flat, it self-intersects.

Reviewer 2

(Major comment) The claim “a sequence of 6 equilateral triangles ...” is incorrect. In Sadowsky’s construction of a nearly optimal paper Möbius band, the relevant configuration consists of 3, rather than 6, equilateral triangles. Nevertheless, the stated ruling length and perimeter are correct, although the computation leading to them is flawed.

For any $\beta > \sqrt{3}$, the domain boundary length 2β strictly exceeds $2\sqrt{3}$. This slack allows us to “open up” the flat equilateral triangle into an embedded, non-self-intersecting Möbius band in $\mathbb{R}^3$ (resembling a highly folded twisted ribbon). By taking a sufficiently fine locally finite G_β -invariant clean triangulation of Σ , and scaling the embedding slightly such that the boundary strictly contracts to a length just below 2β while the internal thickness pushes out to ensure all cross-edges strictly exceed their domain lengths, we can satisfy all conditions of the affine squeeze map.

Reviewer 2

(Major comment) The solution does not justify several nontrivial claims about the properties of the proposed construction. For example, it is not mathematically clear what it means in this context to “‘open up’ the flat equilateral triangle.” The solution also claims that the embedding can be scaled so that “the boundary strictly contracts to a length ...” and that “internal thickness pushes out to ensure all cross-edges strictly exceed their domain lengths,” but these assertions are not accompanied by any mathematical justification. As written, this is only a heuristic description of a construction, rather than a mathematically rigorous argument.

Therefore, a continuous embedding f satisfying all properties exists for all $\beta > \sqrt{3}$, completing the proof that $\sqrt{3}$ is the infimum of the set of realized numbers.

Assessment of Gaps

As requested, we highlight the specific areas where the provided argument remains partial or contains unsupported leaps:

- **Applicability of Schwartz’s Theorem:** Schwartz’s Theorem explicitly applies to smooth or piecewise smooth developable surfaces formed by isometric immersions (i.e., exact paper Möbius bands). The continuous map f constructed in the problem restricts to affine squeeze maps on clean triangles, which strictly *stretches* the cross-segments rather than preserving lengths. The assertion that Schwartz’s inequality $L \geq 2\sqrt{3}w$ transfers immediately to this non-isometric, non-developable polyhedral surface is a non-trivial gap. The concept of “ruling length” and minimal cross-segment bounds requires careful extension.
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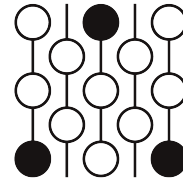
verifying that an affine map strictly satisfies the squeeze condition on *every* individual triangle while maintaining a global topological embedding without self-intersections is left entirely unproven. This step acts merely as a heuristic rather than a constructive mathematical proof.

Review of Submission 2D

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Correctness. The AI did not really solve the problem. This attempt includes its own assessment, which I agree with. The AI gets the argument right in a certain high level sense, but it does not really engage with the problem in a meaningful way. A meaningful engagement would amount to learning the details in my paper and then seeing that essentially the same kinds of ideas work with a different category of maps. Alternately, a really meaningful engagement would involve finding an independent proof.

Novelty. There isn't much here that is novel. The AI is essentially saying that the strategy in my paper could be adapted and made to work somehow. There aren't really proof ideas here, but just sort of "ideas for ideas".

2 Recommendation

I would reject this.

3 Submission with inline comments

I have said everything I want to say above.

Reviewer 3

Problem Statement

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- **Construction for $\beta > \sqrt{3}$:** The second step relies on physical intuition (“open up the flat equilateral triangle”). Finding a rigorous, fine G_β -invariant clean triangulation and explicitly verifying that an affine map strictly satisfies the squeeze condition on *every* individual triangle while maintaining a global topological embedding without self-intersections is left entirely unproven. This step acts merely as a heuristic rather than a constructive mathematical proof.